

In [6]: *#Explain the role of self-attention in transformers with a diagram.*

#Use Hugging Face Transformers to Load 'bert-base-uncased' and print embeddings

#Submit code + explanation.

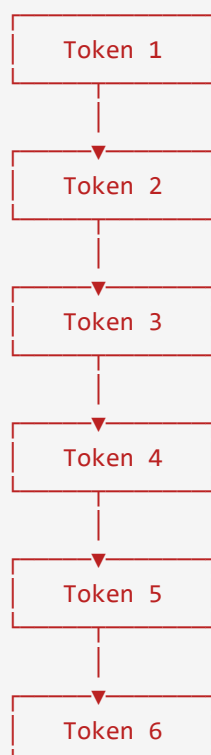
In [7]: '''Self-Attention in Transformers

Role:

Self-attention enables transformers to weigh the importance of each token in a s

Diagram:

Input: [The, cat, sat, on, the, mat]



Each token attends to all other tokens:

[The] <--> [cat] <--> [sat] <--> [on] <--> [the] <--> [mat]

Explanation:

Self-attention computes a weighted sum of all tokens for each token, allowing th

Out[7]: 'Self-Attention in Transformers\nRole:\nSelf-attention enables transformers to weigh the importance of each token in a sequence relative to every other token. This allows the model to capture contextual relationships, regardless of their distance in the input.\n\nDiagram:\nInput: [The, cat, sat, on, the, mat]\n\n

┌──────────┐ \n	Token 1 \n	┌──────────┐ \n	\n
└───▼───┘ \n	Token 2 \n	┌──────────┐ \n	\n
└───▼───┘ \n	Token 3 \n	┌──────────┐ \n	\n
└───▼───┘ \n	Token 4 \n	┌──────────┐ \n	\n
└───▼───┘ \n	Token 5 \n	┌──────────┐ \n	\n
└───▼───┘ \n	Token 6 \n	┌──────────┐ \n	\n

\n\nEach token attends to all other tokens:\n[The] <--> [cat] <--> [sat] <--> [on] <--> [the] <--> [mat] \n\nExplanation:\nSelf-attention computes a weighted sum of all tokens for each token, allowing the model to focus on relevant words for understanding context.'

```
In [8]: # Load BERT and print embeddings for sample text
from transformers import BertTokenizer, BertModel
import torch
```

```
In [9]: # Load pre-trained model and tokenizer
tokenizer = BertTokenizer.from_pretrained('bert-base-uncased')
model = BertModel.from_pretrained('bert-base-uncased')
```

```
tokenizer_config.json: 0%|          | 0.00/48.0 [00:00<?, ?B/s]
```

C:\Users\sadhn\AppData\Roaming\Python\Python312\site-packages\huggingface_hub\file_download.py:143: UserWarning: `huggingface\_hub` cache-system uses symlinks by default to efficiently store duplicated files but your machine does not support them in C:\Users\sadhn\.cache\huggingface\hub\models--bert-base-uncased. Caching files will still work but in a degraded version that might require more space on your disk. This warning can be disabled by setting the `HF\_HUB\_DISABLE\_SYMLINKS\_WARNING` environment variable. For more details, see https://huggingface.co/docs/huggingface_hub/how-to-cache#limitations.

To support symlinks on Windows, you either need to activate Developer Mode or to run Python as an administrator. In order to activate developer mode, see this article: <https://docs.microsoft.com/en-us/windows/apps/get-started/enable-your-device-for-development>

```
warnings.warn(message)
```

```
vocab.txt: 0%|          | 0.00/232k [00:00<?, ?B/s]
```

```
tokenizer.json: 0%|          | 0.00/466k [00:00<?, ?B/s]
```

```
config.json: 0%|          | 0.00/570 [00:00<?, ?B/s]
```

Xet Storage is enabled for this repo, but the 'hf_xet' package is not installed. Falling back to regular HTTP download. For better performance, install the package with: `pip install huggingface\_hub[hf\_xet]` or `pip install hf\_xet`

```
model.safetensors: 0%|          | 0.00/440M [00:00<?, ?B/s]
```

```
In [10]: # Sample text
text = "The cat sat on the mat."
```

```
In [11]: # Tokenize and get input tensors
inputs = tokenizer(text, return_tensors='pt')
```

```
In [12]: # Get embeddings from BERT
with torch.no_grad():
    outputs = model(**inputs)
    embeddings = outputs.last_hidden_state # shape: [batch_size, seq_len, hidden_dim]
```

```
In [13]: print("Embeddings shape:", embeddings.shape)
print("Embeddings for each token:\n", embeddings)
```

```
Embeddings shape: torch.Size([1, 9, 768])
```

```
Embeddings for each token:
```

```
tensor([[-0.3642, -0.0531, -0.3673, ..., -0.3797,  0.5818,  0.4386],
        [-0.3979, -0.2721, -0.6820, ..., -0.3840,  0.9868, -0.3878],
        [-0.3512, -0.0736, -0.0691, ..., -0.6365,  0.6792,  0.5781],
        ...,
        [-0.0415, -0.1055, -0.2808, ...,  0.0057,  0.1986,  0.4328],
        [-0.2354, -0.4875, -0.1631, ...,  0.4162,  0.7360, -0.6378],
        [ 0.6651,  0.0225, -0.4131, ...,  0.1322, -0.2525, -0.3970]])
```

```
In [14]: #Explanation:

#Loads BERT and its tokenizer.
#Tokenizes sample text.
```

```
#Passes tokens through BERT to get embeddings (Last_hidden_state).  
#Prints the shape and values of the embeddings for each token.  
#Each embedding is a vector representing the contextual meaning of a token in th
```

In []: